



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (08/06/2026 – 14/06/2026)

WEEK 4

PROJECT NAME

CHANGE DETECTION

PROJECT DESCRIPTION :

This project aims to detect changes between satellite images captured at different times using Deep Learning and Computer Vision techniques.

GROUP MEMBER :

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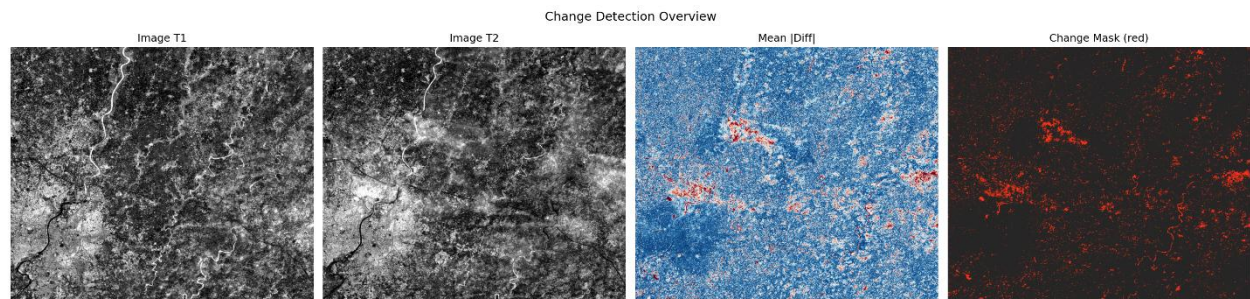
08/06/2026	• Explored the efficient usage of Google Earth Engine in detection.
09/06/2026	• Performed Multivariate Alteration Detection on GeoTiff images of different years. Making labels: Red as New construction and Green as demolition.
10/06/2026	• Getting familier with dataset from GEE.
11/06/2026	• Performing Change detection in images of tiff format downloaded from GEE.
12/06/2026	• Towards enhancement of detected changes so far.
13/06/2026	Holiday (2 nd Saturday)
14/06/2026	Holiday (Sunday)

Next Week Plan	Aimed to make it better visualization and more accurate detection.
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REFERENCE:

- [https://docs.qgis.org/3.44/en/docs/user_manual/working_with_vector/editing_geome try_attributes.html](https://docs.qgis.org/3.44/en/docs/user_manual/working_with_vector/editing_geometry_attributes.html)
- <https://www.earthdata.nasa.gov/learn/data-recipes/detect-environmental-change-using-sar-data-qgis>

SCREEN SHOTS:



Google Earth Engine

Search places and datasets...

Scripts Docs Assets

filter scripts... NEW

Owner (1)
users/rajyavardhan018/bisag_change_detection
This repository is empty.

Writer
No accessible repositories. Click Refresh to check again.

Reader

New Script *

```

12 var before = ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
13   .filterBounds(bisag)
14   .filterDate('2019-01-01', '2019-12-31')
15   .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20))
16   .median()
17   .clip(bisag);
18
19 var after = ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
20   .filterBounds(bisag)
21   .filterDate('2025-01-01', '2025-12-31')
22   .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20))
  
```

Inspector Console Tasks

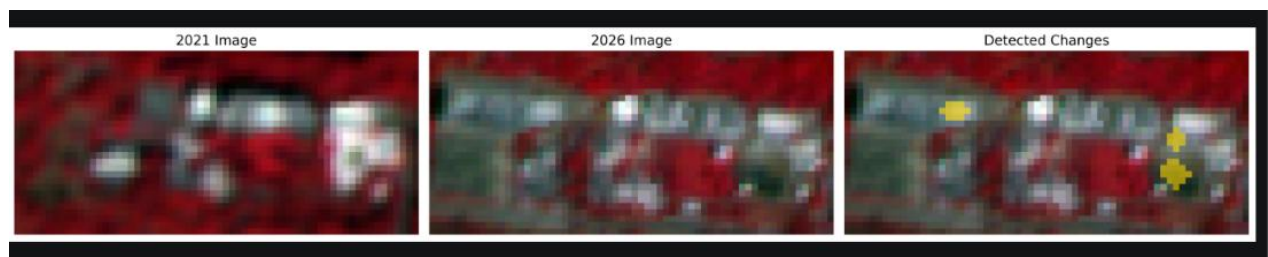
Use print(...) to write to this console.

New Building Area (m²) JSON
Object (1 property) JSON

Demolished Area (m²) JSON
Object (1 property) JSON

geometry (1 poly) Editing: Drag to edit vertices. Delete Exit

Layers Map Satellite



Major Change Visualizations

Before (Reference: 2021)



Original image for the base year 2021.

After (Target: 2022)



Aligned image for comparison year 2022.

Detected Major Changes



Change zones highlighted in red overlay (noise filtered).

